

LC A2

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

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Warning

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.



Warning

The use of this product requires experience with and knowledge of the product. Persons with reduced physical, sensory or mental capabilities must not use this product, unless they are under supervision or have been instructed in the use of the product by a person responsible for their safety. Children must not use or play with this product.

1. Symbols used in this document



Warning

If these safety instructions are not observed, it may result in personal injury.



Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.



Note

Notes or instructions that make the job easier and ensure safe operation.

The safety instructions in the manuals for the lifting station and the appliances connected must also be observed.

2. Product introduction

The LC A2 alarm device is designed for stopping washing machines or dishwashers to prevent flooding.

3. Applications

The LC A2 is intended for the monitoring of pumping stations and small lifting stations.

A level switch, installed in the appliance to be monitored, will be activated for example in case of overflow due to pump failure or clogging. The unit will sound alarm and activate an integrated relay contact to switch off the power supply to the washing machine connected. The function, which is preset, prevents flooding by stopping the washing machine pump.

3.1 Construction

The LC A2 consists of a plug housing with integrated socket. The socket can be used as power supply for other appliances and for switching on/off appliances in case of fault.

The unit is mains-powered by means of a short-circuit-proof safety isolating transformer. The voltage for the control relay (normally open contact) is below 24 V.

The LC A2 has a potential-free signal output.

4. Installation

Warning



The protective earth (PE) of the power outlet must be connected to the protective earth of the product. The plug must have the same PE connection system as the power outlet.

Warning



The installation must be fitted with a residual current device (RCD) with a tripping current less than 30 mA.

Warning



The product must be connected to an external main switch with a minimum contact gap of 3 mm (0.12 inch) in all poles.

Warning



Unplug the unit before opening it.

1. Install the level switch. See examples on page 6.
2. Remove the four screws at the back of the LC A2 and lift off the cover carefully.
3. Disconnect wires, if any, to the buzzer.
4. Remove the cover.
5. Lead the level switch cable through a suitable cable gland and connect the conductors to the terminal block.
6. Tighten the cable gland by hand.
7. Follow the same procedure for the alarm relay.
8. Connect the buzzer by means of the connector.
9. Fit the cover.

Caution Make sure that wires are not pinched.

10. Fit and tighten the four screws.

5. Examples of systems

See page 6.

Caution Observe limitations, if any, in the use of pumps or lifting stations.

6. Level switches

6.1 Float switch

Float switch, type MS2, for installation in tanks and pits. It is especially suitable for Liftaway B, Liftaway C 100, Unilift and PUST.



Fig. 1 Float switch

91427145: Float switch with 5 m cable.

91427146: Float switch with 10 m cable.

91427147: Float switch with 20 m cable.

6.2 Vertical level switch

Level switch for installation in the cover of small lifting stations and tanks. It is especially suitable for Sololift2 C-3, Sololift+ C-3 and Liftaway C.



Fig. 2 Vertical level switch

97775337: Vertical float switch with 10 m cable.

TM01 6982 3999

TM05 0630 1311

7. Electrical connection

Figure 3 shows the position of the terminal blocks for the control relay (connection of level switch) and the alarm relay.

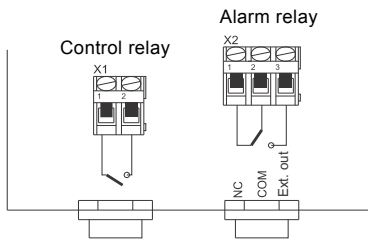


Fig. 3 Alarm relay

The socket function can be set at the terminal block. See fig. 4.

- Cont.:
The socket is used as a feed-through socket.
- NO (normally open contact):
The socket will be switched on in case of fault.

- NC (normally closed contact):
The socket will be switched off in case of fault. This is the default setting for stop of a washing machine.

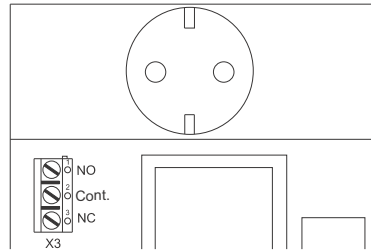


Fig. 4 Terminal block, socket function

By connecting several LC A2 units, a corresponding number of appliances can be switched on/off by means of one control relay (level switch). See example in fig. 5.

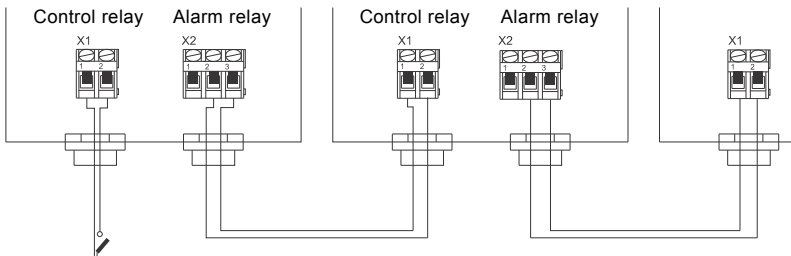


Fig. 5 Connection of several LC A2 units

It is also possible to lock the LC A2 to disable automatic restarting. In case of alarm, the LC A2 must be pulled out of the socket. See example in fig. 6.

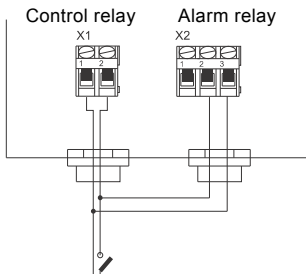


Fig. 6 Locked LC A2

8. Technical data

Enclosure

Flame-retardant polystyrene according to UL 94 V-0 or 94 V-2, impact-resistant, bicolour, with integrated socket (can be switched on or off), two cable glands with cable relief, 2 x M12 x 1.5.

Power supply

230 V, 50 Hz, short-circuit-proof safety isolating transformer.

No-load power consumption: 0.2 VA.

Power consumption in case of alarm: approx. 2 VA.

Input

Potential-free normally open contact, min. 24 V, 75 mA.

Output

Potential-free contact, 3 A, 250 VAC or 3 A, 30 VDC, short-term 5 A.

In case of alarm, the socket can be switched on/off (15 A / 250 VAC, 1-pole). It can also be used as feed-through socket.

An acoustic alarm signal is possible by means of a piezoelectric buzzer with a tone sequence signal of 97 dB / 1 m.

Ambient temperature

0-40 °C.

Dimensions

140 x 60 x 90 mm.

9. Disposal

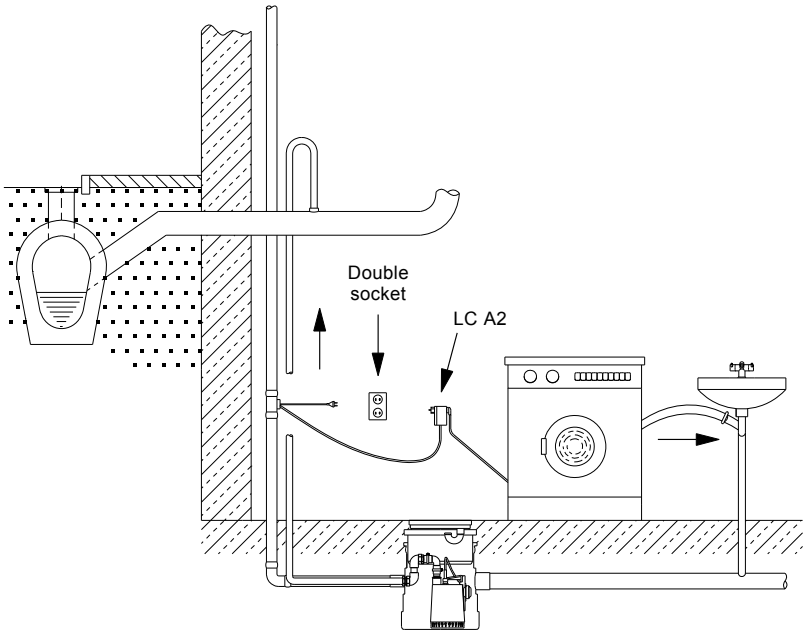
This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.



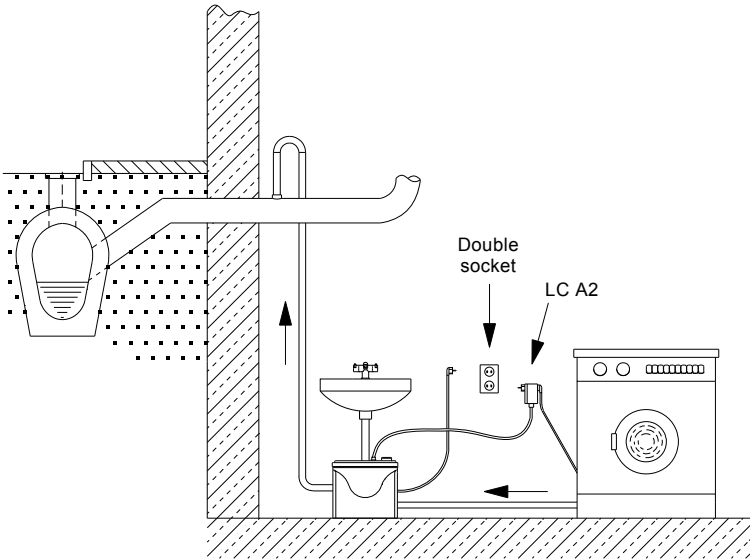
The crossed-out wheeled bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local

waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.



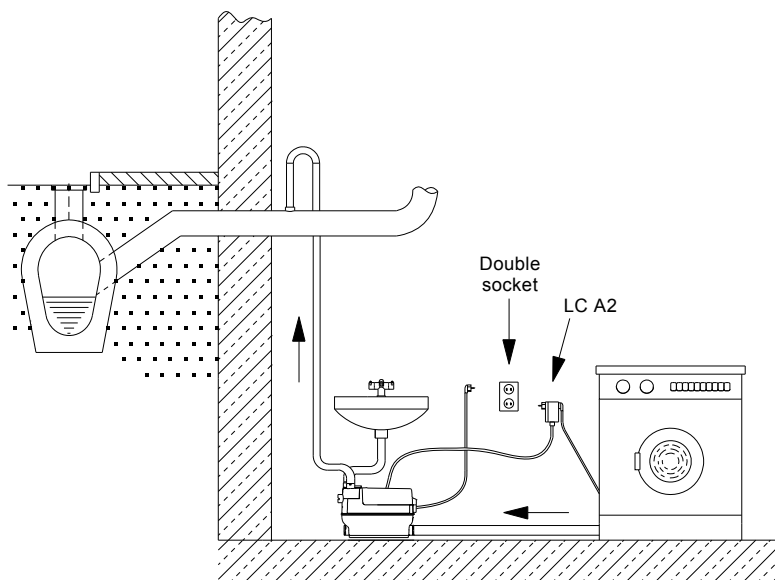
TM05 0513 1311

Fig. 1 Installation with Liftaway B



TM05 0514 1311

Fig. 2 Installation with Liftaway C



TM05 0550 1211

Fig. 3 Installation with Sololift2 C-3



TM05 0634 1311

Fig. 4 Vertical level switch in Sololift2 C-3

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